

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. We prove a finite-time hypocoercive observability estimate and closed range for the linearized balance operator. Exact localized finite-volume cumulants then yield a tight joint density–contact Gaussian process on a nuclear dual space. The collision random measure uses control qA_f and integrand $\Delta p + \psi$, so the biased intensity is counted once. An independent exactly balanced quadratic recovery establishes Mosco convergence and twice epi-differentiability of the action on the Cameron–Martin domain $\text{Ran } \Sigma^{1/2}$.

Platform identifier: HS-JOINT-GAUSSIAN-MOSCO-v17.

Controlling revision: ROUND17-REFEREE-POSITIVE-CLOSURE.

Registered proof source: B3_CUMULANT_MOSCO_PROCESS.tex.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: JOINT CUMULANTS, OBSERVABILITY, AND QUADRATIC MOSCO LIMITS

The Gaussian field is constructed first from exact finite-volume cumulants. Only afterwards is its Cameron–Martin form identified with the second variation of the density–contact action. The collision noise convention is fixed once and for all: the isonormal measure has control qA_f , and the integrand is $\Delta p + \psi$; equivalently one may use control A_f and integrand $\sqrt{q}(\Delta p + \psi)$, but never both factors simultaneously.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 60F10, 82C40; Secondary 49L25, 76P05.

Key words and phrases. Boltzmann equation, kinetic Hamiltonian, collision flow, cotangent field, fluctuating Boltzmann equation.

1.1. A fixed nuclear triple and the linearized balance. Let $\mathcal{S}$ be the projective intersection of weighted Sobolev spaces $H_x^m H_v^m$ with Gaussian velocity weights and compatible endpoint traces, and let

$$\mathcal{S} \subset \mathcal{H} \subset \mathcal{S}'$$

be the resulting nuclear Gelfand triple. For collision tests use the corresponding symmetric nuclear space $\mathcal{S}_c$ on (t, x, v, v_*, ω) modulo $(v, v_*, \omega) \sim (v_*, v, -\omega)$. Put $\mathcal{E} = \mathcal{S} \oplus \mathcal{S}_c$.

At a regular biased path (f, qA_f) , let $\mathcal{L}_t$ be the linearized transport-collision operator. A tangent pair (h, K) satisfies

$$\partial_t h + v \cdot \nabla_x h = \Delta^* K, \quad K = q DA_f[h] + A_f k,$$

where k is the relative collision-intensity tangent.

Theorem 1.1 (Finite-time kinetic observability). *On the orthogonal complement of mass, momentum, energy, and the exact balance gauge, there are $c, C > 0$ such that every smooth backward solution*

$$-\partial_t p - v \cdot \nabla_x p - \mathcal{L}_t^* p = F, \quad p(T) = p_T,$$

satisfies

$$\|p\|_{L^2(0,T;H_{x,v,w}^1)}^2 + \|p(0)\|_{L_w^2}^2 + \|p(T)\|_{L_w^2}^2 \leq C \left(\|F\|_{L^2(0,T;H_w^{-1})}^2 + \int |\Delta p|^2 q dA_f \right).$$

Consequently the linearized balance operator has closed range on the declared weighted spaces and admits a bounded right inverse on its range.

Proof. Freeze coefficients on a time partition and Fourier transform in x . For the zero spatial mode, the biased collision operator is a bounded perturbation of the Maxwellian linearized operator; the spectral gap on the complement of collision invariants gives velocity coercivity. For nonzero spatial modes, the commutator of $iv \cdot k$ with a bounded velocity multiplier transfers the collision coercivity to the hydrodynamic component. The standard modified energy

$$E_k(p) = \|p\|_{L_w^2}^2 + \eta \Re \langle A_k p, p \rangle$$

then obeys

$$-\frac{d}{dt} E_k(p) + c(\|p_\perp\|_{H_v^1}^2 + \frac{|k|^2}{1+|k|^2} \|p_\parallel\|^2) \leq C \|F\|_{H^{-1}}^2.$$

The constants are uniform on the small regular source chart because f and q are bounded above and below relative to the reference Maxwellian and have uniform weighted derivatives. Integrating, summing in k , and treating the finite-dimensional invariant block by the endpoint conditions gives the displayed estimate. Approximation of the nonautonomous coefficients by the time partition and Gronwall's lemma pass to the true equation. The closed range and bounded inverse follow from the Hilbert closed-range theorem. $\square$

1.2. Exact finite cumulants and process tightness. For a test pair $\Theta = (p, \psi) \in \mathcal{E}$, let $X_\varepsilon(\Theta)$ be the centered density/contact observable under the regular biased microscopic law. B2's normally convergent marked pressure, combined with B1's exact shell saddle when preparation is imposed, gives finite-volume cumulants $\kappa_{r,\varepsilon}$.

Lemma 1.2 (Localized cumulant estimate). *For every Hilbertian seminorm $|\cdot|_m$ generating the nuclear topology, there are $m' > m$ and constants C_r such that for tests supported in a time interval $I \subset [0, T]$,*

$$|\kappa_{r,\varepsilon}(\Theta_1, \dots, \Theta_r)| \leq C_r \mu_\varepsilon^{1-r/2} (|I| + \mu_\varepsilon^{-1}) \prod_{j=1}^r |\Theta_j|_{m'}$$

for $r \geq 2$. The estimate is uniform over all deterministic intervals, including intervals shorter than μ_ε^{-1} , and after exact microcanonical centering.

Proof. Differentiate the exact marked trajectory-cluster expansion with each source localized to I . Every connected diagram contributing to an r -th cumulant has one chronologically first marked vertex; its time integral gives $|I|$. If two or more marks lie in the same microscopic contact, the diagonal term contributes one factor μ_ε^{-1} instead. The remaining tree integrations are bounded by the fixed-horizon B2 factorial majorant. After multiplying by the central-limit normalization $\mu_\varepsilon^{-r/2}$, connectedness leaves the factor $\mu_\varepsilon^{1-r/2}$. The B1 source-dependent saddle is analytic with uniform derivatives, so conditional cumulants are obtained by the finite Schur-complement differentiation and obey the same estimate. $\square$

Theorem 1.3 (Joint density–contact functional central limit theorem). *The centered field*

$$Z_\varepsilon = \sqrt{\mu_\varepsilon}((\pi^\varepsilon, \Gamma^\varepsilon) - (f, qA_f))$$

converges in $C([0, T]; \mathcal{S}') \times D([0, T]; \mathcal{S}'_c)$ to the unique Gaussian solution of the linearized balance equation. Its collision martingale is the isonormal integral

$$M(p, \psi) = \int (\Delta p + \psi) d\mathbb{W}_{qA_f},$$

and hence

$$\mathbf{E}[M(p, \psi)M(\tilde{p}, \tilde{\psi})] = \int (\Delta p + \psi)(\Delta \tilde{p} + \tilde{\psi}) q dA_f.$$

The initial covariance is the B1 constrained Schur complement and is propagated by the nonautonomous linearized evolution together with all density–contact cross terms.

Proof. For every finite set of tests, Lemma 1.2 makes all cumulants of order at least three vanish and identifies the limiting second cumulant by differentiating the B2 Hamiltonian. This gives finite-dimensional Gaussian convergence with the displayed, singly weighted collision bracket.

For tightness, choose a Hilbertian seminorm $|\cdot|_m$ for which the embedding of the next stronger dual space is Hilbert–Schmidt. Apply the second- and

fourth-cumulant estimates to every dyadic interval and every vector of an orthonormal determining family. Summation using the Hilbert–Schmidt embedding yields

$$\mathbf{E}\|Z_\varepsilon(t) - Z_\varepsilon(s)\|_{-m}^4 \leq C(|t - s| + \mu_\varepsilon^{-1})^2.$$

The probability of a jump larger than η is bounded by the marked contact exponential moment and tends to zero. Kolmogorov–Aldous tightness followed by Mitoma’s theorem gives tightness in the stated nuclear dual path space. The exact microscopic balance passes to the limit and identifies the unique Gaussian martingale problem by Theorem 1.1. $\square$

1.3. Second-order action and Mosco identification. Write a finite-action collision flow as $(q + ho)A_{f+h}$ near the exposed path. The scalar entropy has the uniform expansion

$$\ell(q + ho) = \ell(q) + \rho \log q + \frac{\rho^2}{2q} + R_q(\rho), \quad |R_q(\rho)| \leq C|\rho|^3$$

when q stays in a compact subset of $(0, \infty)$.

Theorem 1.4 (Quadratic recovery and Mosco convergence). *Let I be the $B2$ joint density–contact action and x_* a regular exposed path. The rescaled convex functionals*

$$I_\eta(z) = \frac{I(x_* + \eta z) - I(x_*) - \eta \langle DI(x_*), z \rangle}{\eta^2/2}$$

Mosco-converge on the balanced tangent Hilbert space to

$$\mathcal{Q}(z) = \|\Sigma^{-1/2}z\|^2, \quad D(\mathcal{Q}) = \text{Ran } \Sigma^{1/2}.$$

The limsup is supplied by an explicit positive, exactly balanced quadratic recovery; it does not use the first-order LDP recovery theorem.

Proof. For the liminf, take $z_\eta \rightharpoonup z$ with bounded I_η . Entropy coercivity gives boundedness of the relative collision perturbations in $L^2(qA_f)$, and the exact balance gives weak compactness of the density perturbations. Taylor’s formula with truncation at $|\rho| \leq \eta^{-1/3}$ and entropy control of the complement yields the quadratic lower bound. Closedness of the balance range identifies its domain with the quotient tangent space.

For the limsup, first take a smooth tangent $z = (h, K)$ in the range of the linearized balance. Use the bounded right inverse from Theorem 1.1 to construct a second-order correction $(h^{(2)}, K^{(2)})$ cancelling the nonlinear balance defect of $x_* + \eta z$. Set

$$f_\eta = f + \eta h + \eta^2 h^{(2)}, \quad \Gamma_\eta = qA_f + \eta K + \eta^2 K^{(2)}.$$

After truncating the smooth tangent, f_η and Γ_η remain strictly positive. Direct substitution into the entropy expansion gives $I_\eta(z_\eta) \rightarrow \mathcal{Q}(z)$. Density of smooth range vectors in $\text{Ran } \Sigma^{1/2}$ and a diagonal argument complete the limsup. Convex duality of the limiting Gaussian pressure identifies the quadratic form with Σ^{-1} . $\square$

Theorem 1.5 (Round-seventeen B3 closure). *On the regular short-time phase chart, the balance operator has closed range, the finite microscopic cumulants generate a tight joint density–contact Gaussian process with correctly normalized collision noise, and the joint action is twice epi-differentiable with Cameron–Martin domain $\text{Ran } \Sigma^{1/2}$.*

Proof. Combine Theorems 1.1, 1.3, and 1.4, using Lemma 1.2. □

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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